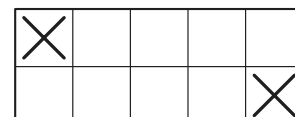
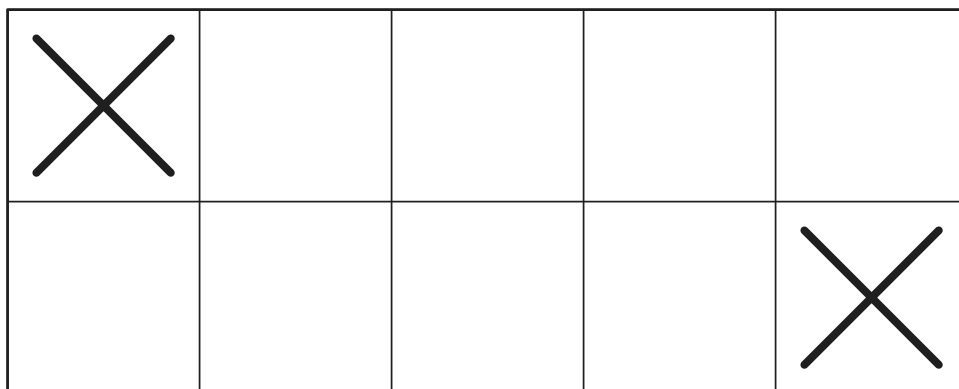
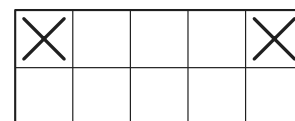
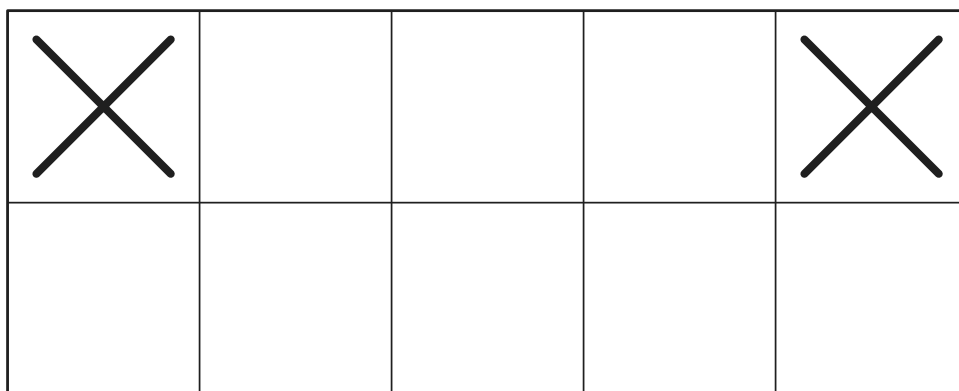


Problem 1: A domino covers two squares. On each of the boards A to F, two squares are crossed out and may not be covered. Try to cover the other squares with dominoes: every square covered, no overlaps, nothing sticking out. When you succeed, draw your cover on the small board beside it. When you decide that it cannot be done, cross out the small board. Look at your six results side by side.

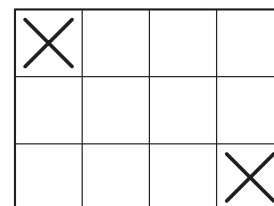
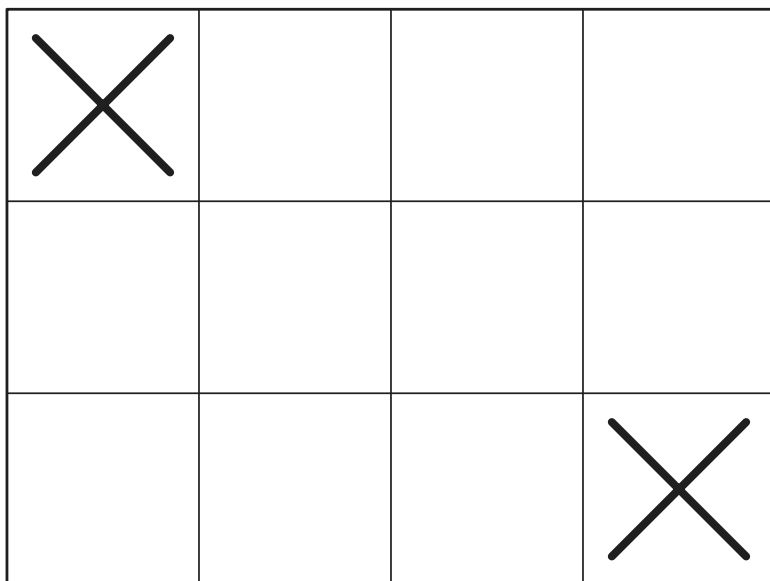
A



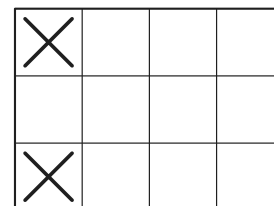
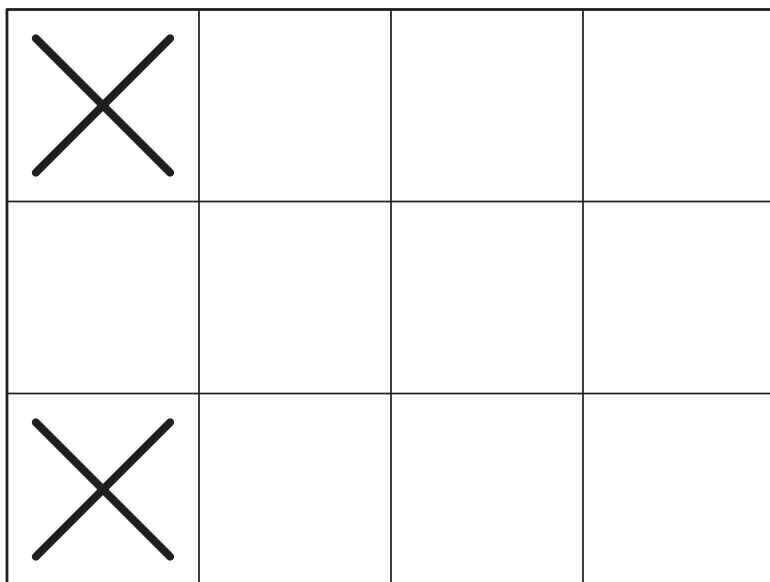
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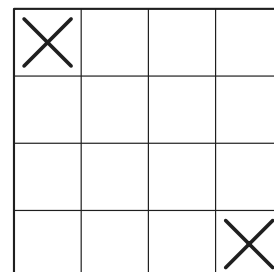
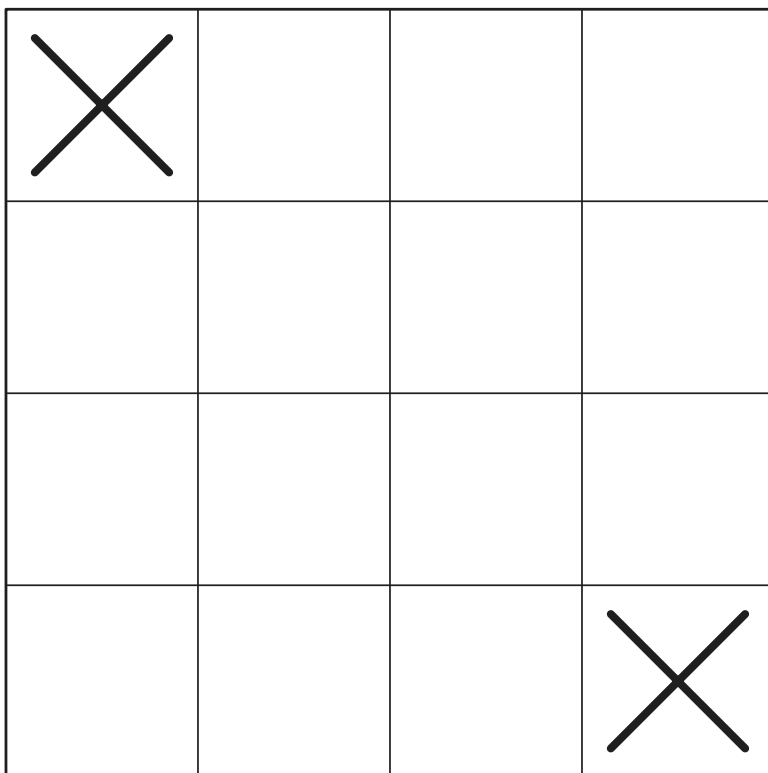
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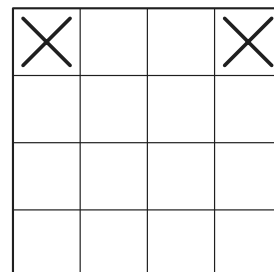
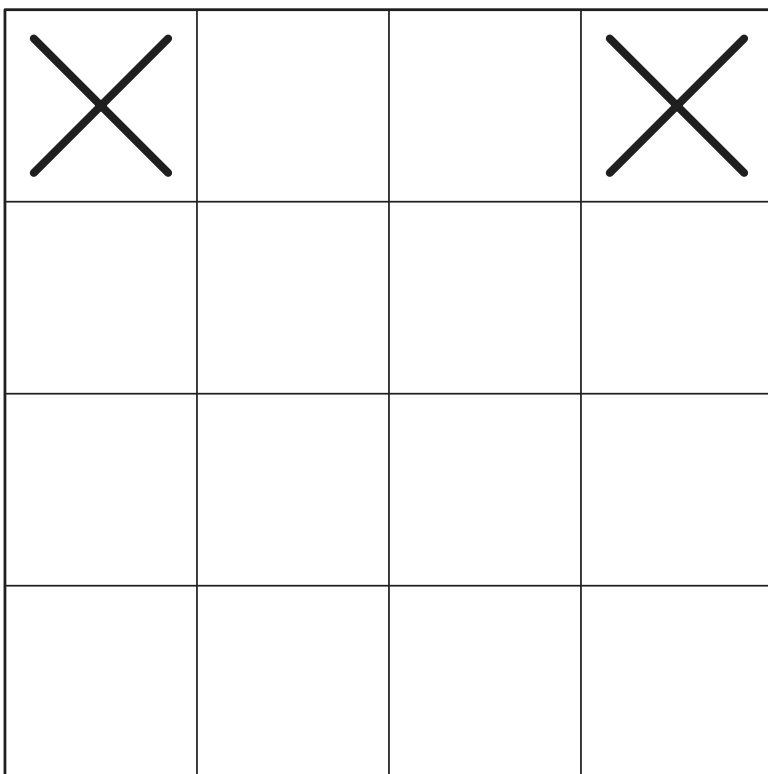
D



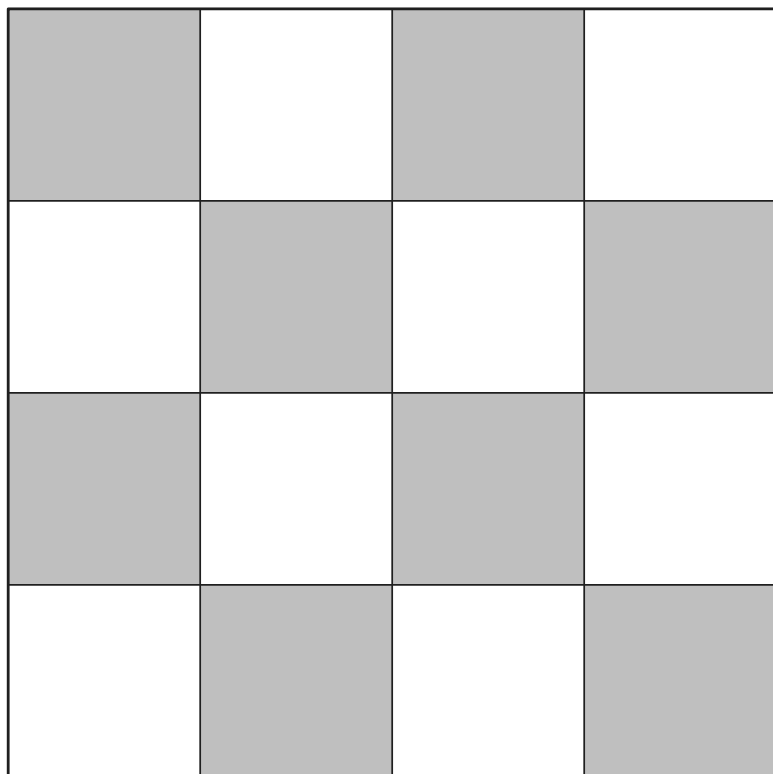
E



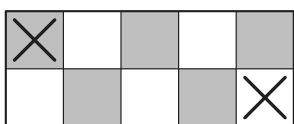
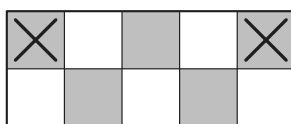
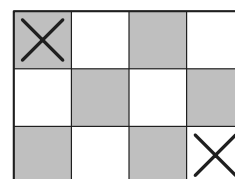
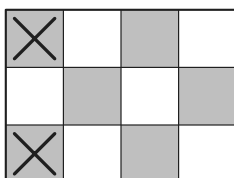
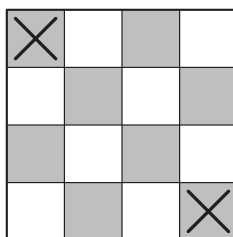
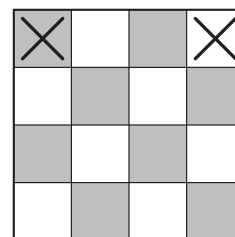
F



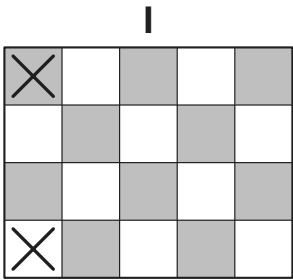
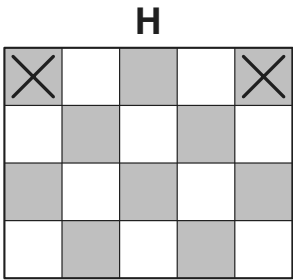
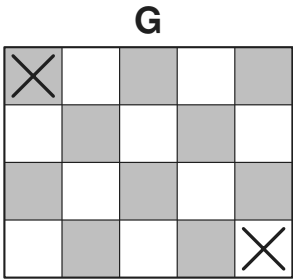
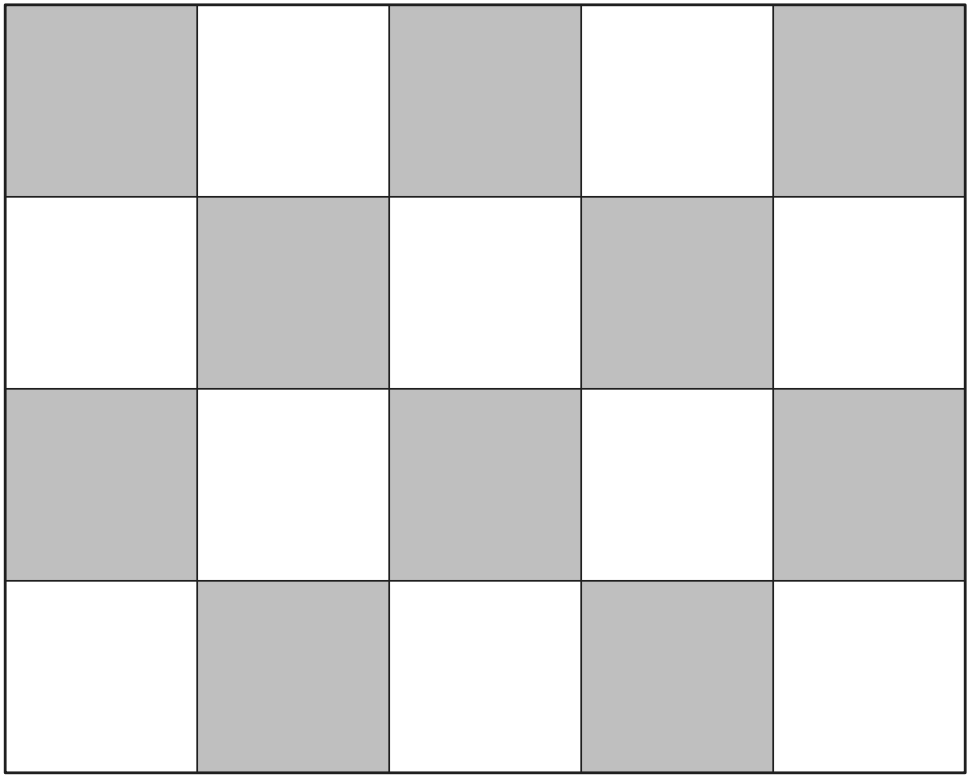
Problem 2: This 4-by-4 board is shaded like a checkerboard. Put one domino anywhere on it and look at the colours of the two squares under it. Move it to five other places and look each time. Now, on the shaded pictures of boards A to F, count the dark squares and the light squares that are not crossed out. Write the numbers in the table. In the last column, write yes or no: was that board covered in Problem 1?



	dark	light	covered in Problem 1?
A			
B			
C			
D			
E			
F			

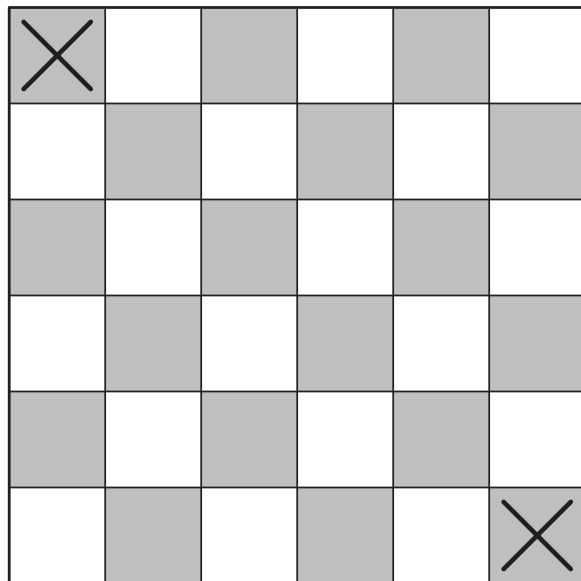
A**B****C****D****E****F**

Problem 3: Boards G, H and I are the same 4-by-5 board with two squares crossed out, as shown in the small pictures. Before you place any domino: count the dark and the light squares in each small picture, write them in the table, and write your prediction, yes or no. Then put the X squares on the big board as in picture G and test it with dominoes. Write what happened in the last column. Do the same for H and I.



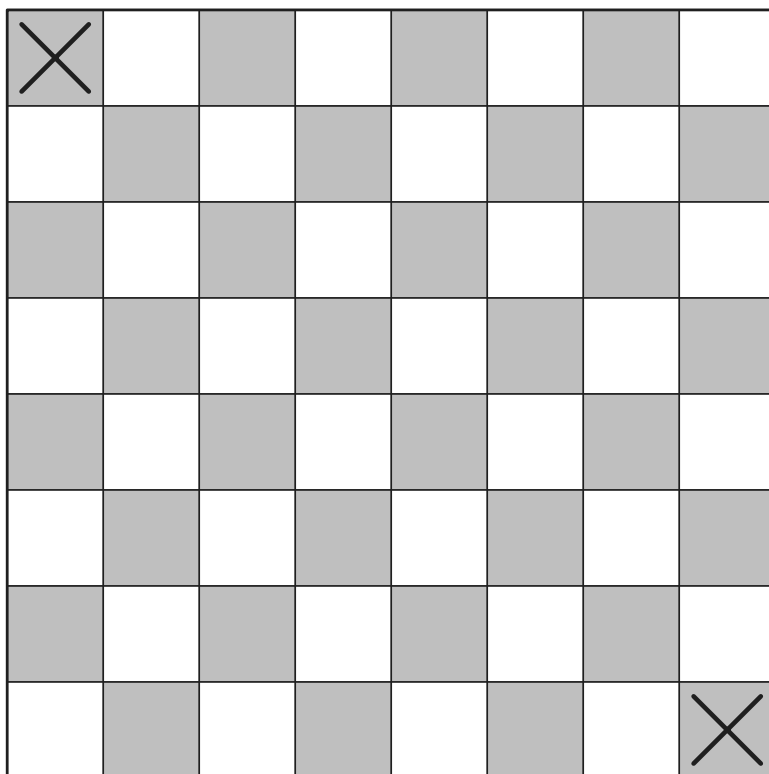
	dark	light	prediction	result
G				
H				
I				

Problem 4: On the 6-by-6 board and on the 8-by-8 board, two opposite corners are crossed out. Try to cover each board by drawing dominoes in pencil. Count the dark squares and the light squares that are not crossed out and write the numbers in the boxes. If a board cannot be covered, tell the adult why.



dark

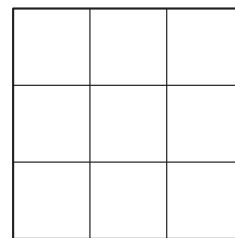
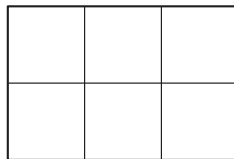
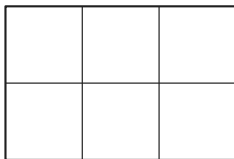
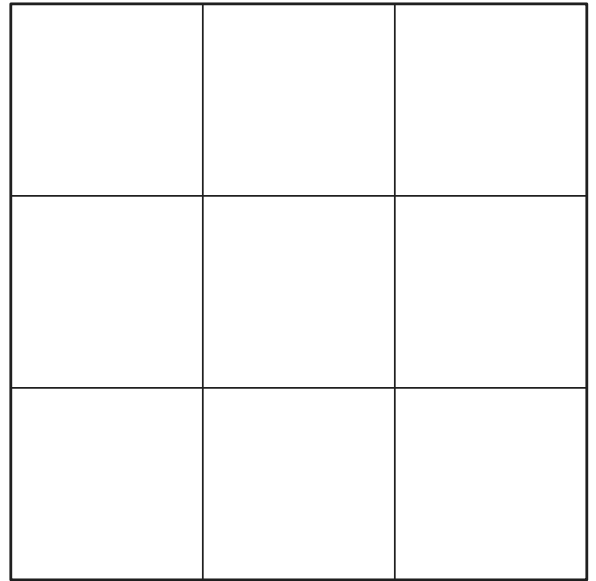
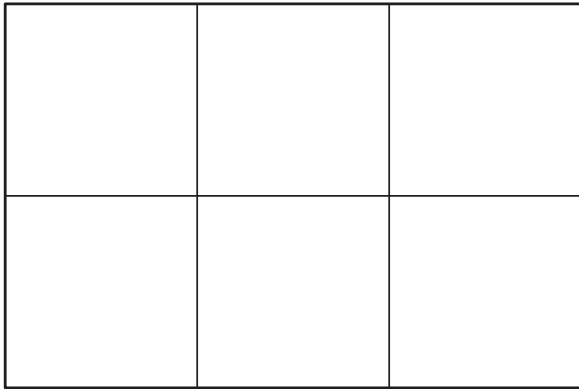
light



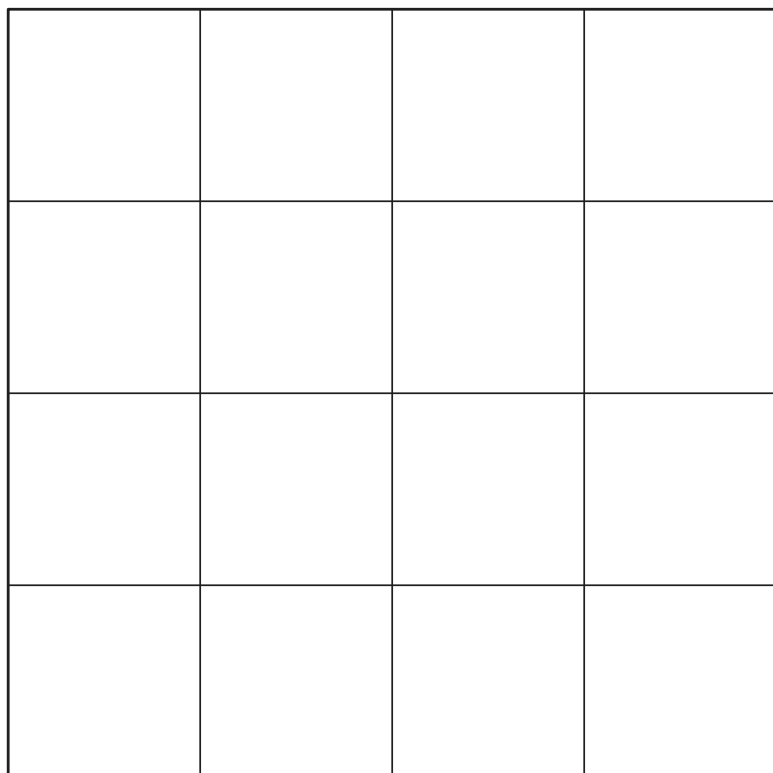
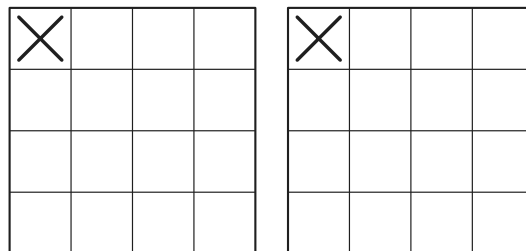
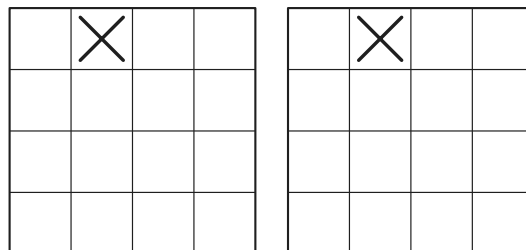
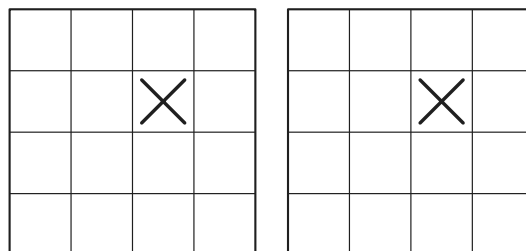
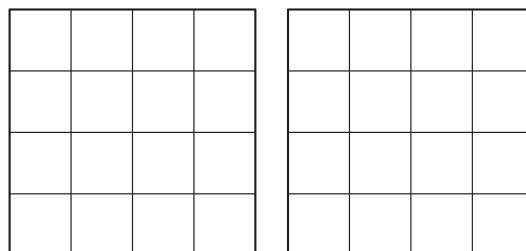
dark

light

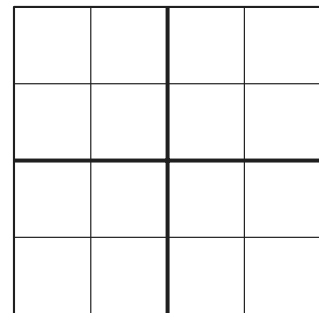
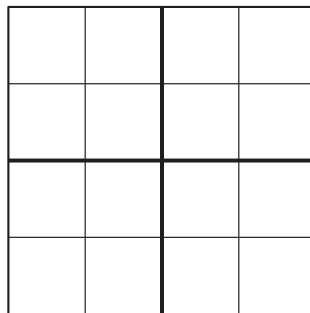
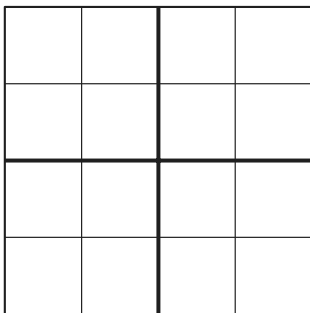
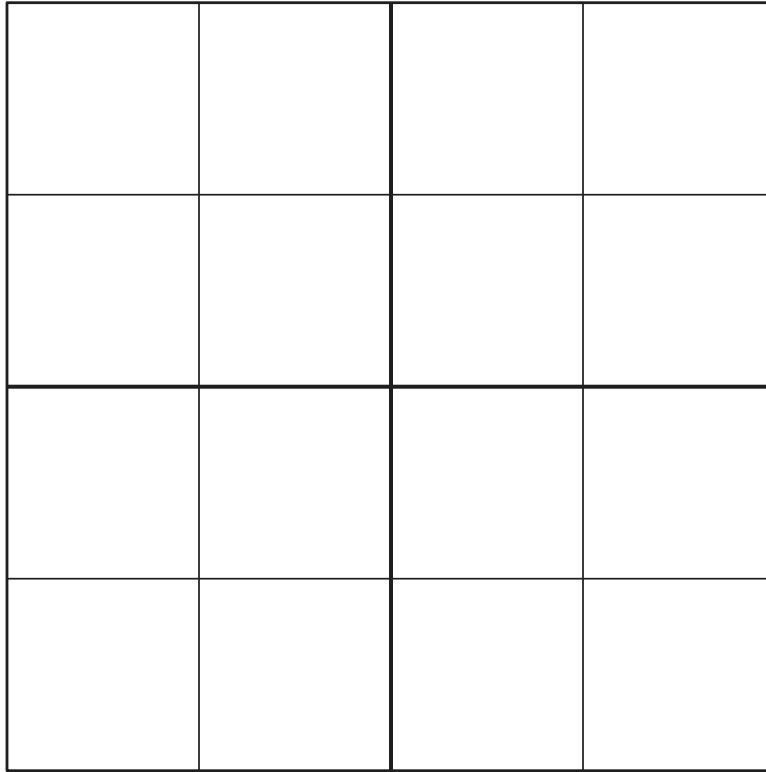
Problem 5: An L-piece covers three squares in an L shape. Cover the 2-by-3 board with two L-pieces and draw the cover on the first small board. Without turning the board, find a different cover and draw it on the second small board. Now try to cover the 3-by-3 board with three L-pieces. Draw it on the third small board, or cross that small board out and tell the adult why it cannot be done.



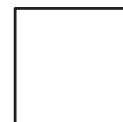
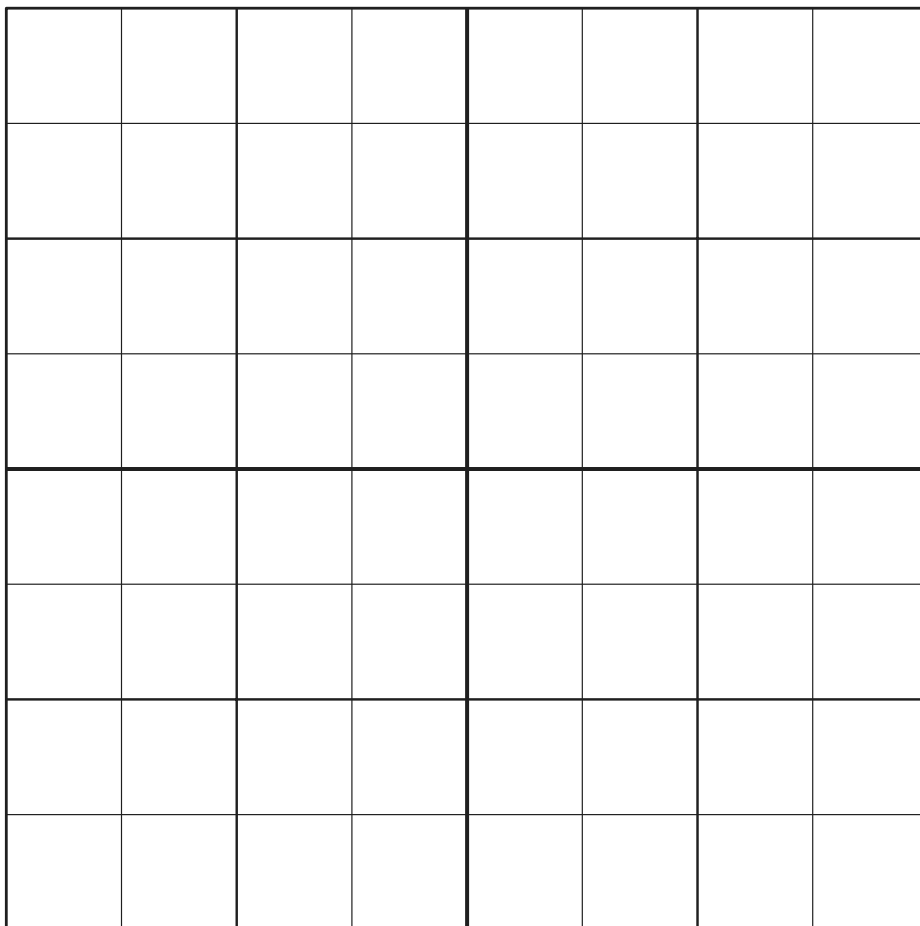
Problem 6: Put the X square on the corner of the 4-by-4 board shown in picture A. Cover the other 15 squares with five L-pieces and draw your cover on picture A. Then look for a second cover of the same board that looks different, without turning the board. Draw it on the second picture A, or cross that picture out if there is none. Do the same for the edge square in picture B, the inside square in picture C, and a square of your own choice, which you mark on picture D.

**A****B****C****D**

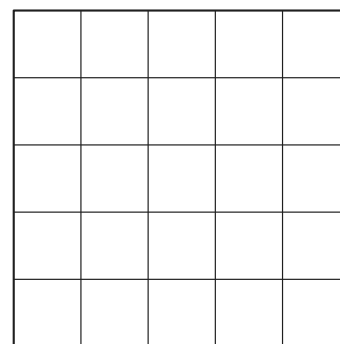
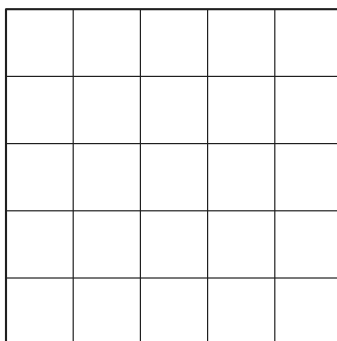
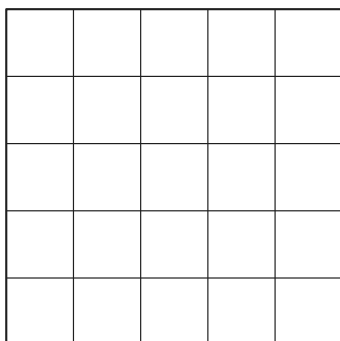
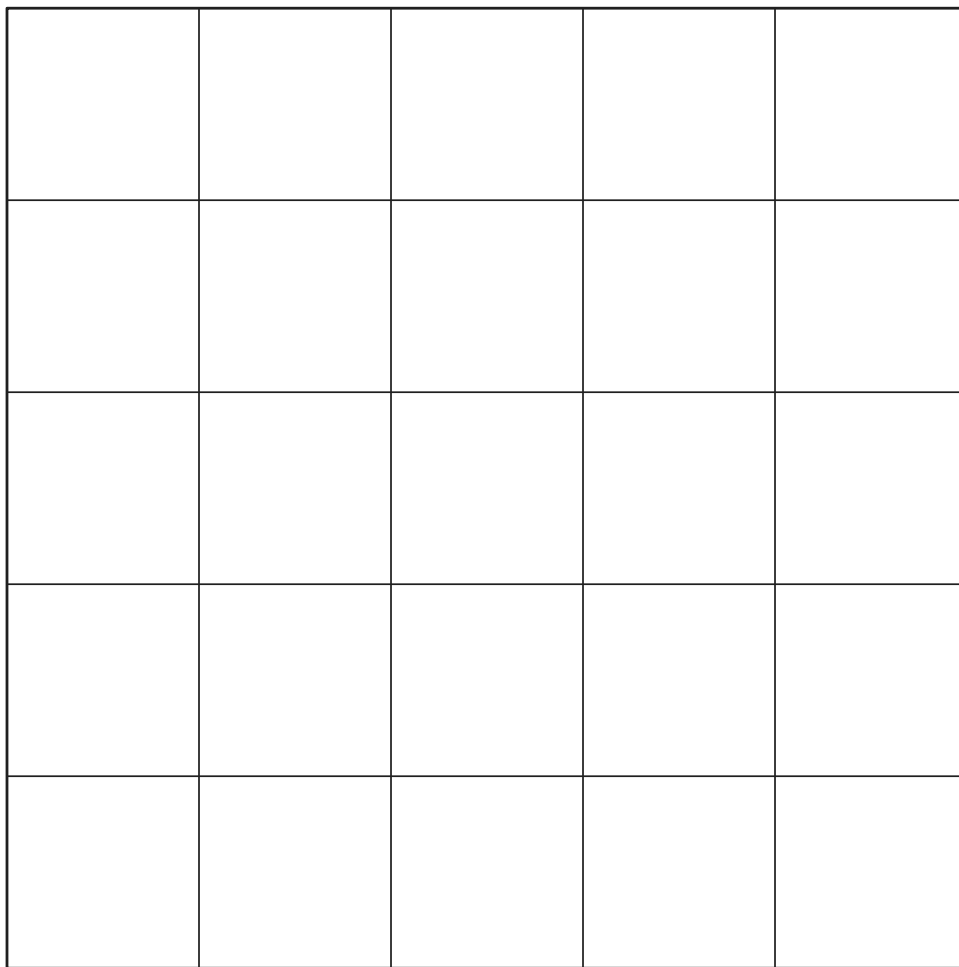
Problem 7: The thick lines cut this 4-by-4 board into four 2-by-2 quarters. Put the X square on any square. Place one L-piece in the middle of the board so that it covers one square in each of the three quarters that do not contain the X square. Finish the cover with four more L-pieces. Draw the result on the first small board and shade the middle L-piece. Do this for two more X positions in different quarters, on the second and third small boards.



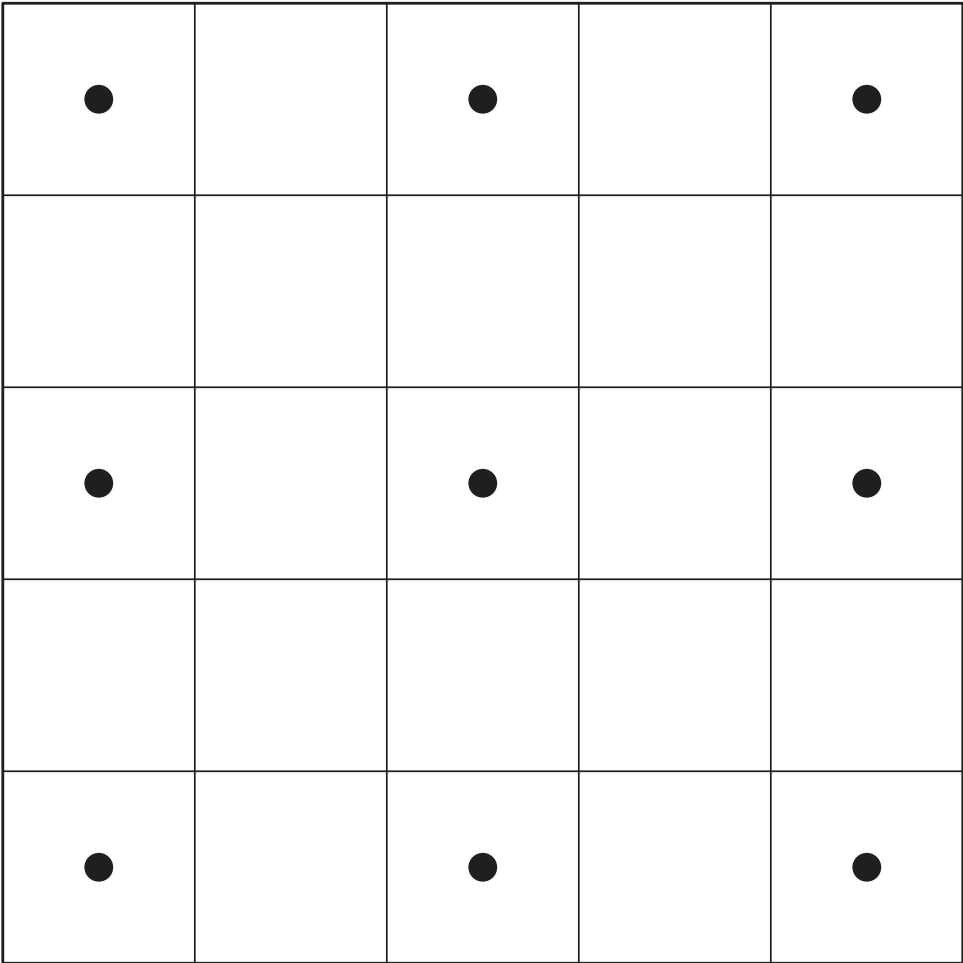
Problem 8: Cross out any one square of the 8-by-8 board. Draw one L-piece in the middle of the board that covers one square in each of the three big quarters that do not contain your crossed-out square. Now each big quarter is a 4-by-4 board with one square missing. Do the same trick inside each big quarter, and then inside each 2-by-2 block. Count the L-pieces you drew and write the number in the box.



Problem 9: Put the X square on the centre square of the 5-by-5 board and cover the other 24 squares with eight L-pieces. Draw the cover on the first small board. Now put the X square on a corner, then on the middle square of the top edge, then on the square next to the corner along the top edge. Each time, draw a tick or a cross in the result grid in the square where the X was. Try more squares until you can see which squares work. Then take the dotted board. Put one L-piece anywhere on it and count the dots it covers. Try six places. Write in the box the most dots that eight L-pieces could cover.



result grid



dotted board